Bing-yue Wu

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Education

Arizona State University 2023 – Present
Ph.D. in Electrical Engineering *Tempe, Arizona*

National Taiwan University of Science and Technology (Taiwan Tech, NTUST) 2021 – 2023
M.S. in Electrical Engineering *Taipei, Taiwan*

National Taiwan University of Science and Technology (Taiwan Tech, NTUST) 2016 – 2020
B.S. with major in Electrical Engineering and minor in Computer Science and Information Engineering *Taipei, Taiwan*

Publications

Z. Jiang, Q. Wu, **B.-Y. Wu**, and J. Zhang, "**CAPO: Certification-Guided Agentic Workflow for Physical Design Parameter Optimization**", *in Proc. GLSVLSI*, 2026.

B.-Y. Wu, C.-T. Ho, H. Yang, C.-C. Chang, A. B. Akkur, B. Khailany, and V. A. Chhabria, "**Special Research Session: Toward Agentic Solution for DRC Challenges in Digital VLSI Design**", *in Proc. DAC*, 2026. [GitHub]

B.-Y. Wu, A. Dey, A. Rovinski, and V. A. Chhabria, "**Focus Session: Large Language Models in Physical Design: From Data Generation to Intelligent Agents**", *in Proc. DATE*, 2026.

B.-Y. Wu and V. A. Chhabria, "**DALI-PD: Diffusion-based Synthetic Layout Heatmap Generation for ML in Physical Design**", *in Proc. ASP-DAC*, 2026. [GitHub]

V. A. Chhabria, A. Ghose, V. Gopalakrishnan, A. B. Kahng, S. Kundu, Y. Liu, Z. Wang, and **B.-Y. Wu**, "**IEEE DATC RDF-2025: Enabling an EDA Research Ecosystem**", *in Proc. ICCAD*, 2025.

B.-Y. Wu, U. Sharma, A. Rovinski, and V. A. Chhabria, "**OpenROAD Agent: An Intelligent Self-Correcting Script Generator for OpenROAD**", *in Proc. ICLAD*, 2025. [GitHub]

V. A. Chhabria, V. Gopalakrishnan, A. B. Kahng, S. Kundu, Z. Wang, **B.-Y. Wu**, and D. Yoon, "**Strengthening the Foundations of IC Physical Design and ML EDA Research**", *in Proc. ICCAD*, 2024.

V. A. Chhabria, **B.-Y. Wu**, U. Sharma, K. Kunal, A. Rovinski, and S. S. Sapatnekar, "**Generative Methods in EDA: Innovations in Dataset Generation and EDA Tool Assistants**", *in Proc. ICCAD*, 2024.

B.-Y. Wu, R. Liang, G. Pradipta, A. Agnesina, H. Ren, and V. A. Chhabria, "**2024 ICCAD CAD Contest Problem C: Scalable Logic Gate Sizing Using ML Techniques and GPU Acceleration**", *in Proc. ICCAD*, 2024. [GitHub]

U. Sharma*, **B.-Y. Wu***, S. R. D. Kankipati, V. A. Chhabria, and A. Rovinski, "**OpenROAD-Assistant: An Open-Source Large Language Model for Physical Design Tasks**", *in Proc. MLCAD*, 2024. [GitHub]

V. Gopalakrishnan, **B.-Y. Wu**, and V. A. Chhabria, "**ML-INSIGHT: Machine Learning for Inrush Current Prediction and Power Switch Network Improvement**", *in Proc. ISLPED*, 2024.

B.-Y. Wu, U. Sharma, S. R. D. Kankipati, A. Yadav, B. K. George, S. R. Guntupalli, A. Rovinski, and V. A. Chhabria, "**EDA Corpus: A Large Language Model Dataset for Enhanced Interaction with OpenROAD**", *in Proc. LAD*, 2024. **(Best Paper Nominated)** [GitHub]

V. A. Chhabria, W. Jiang, A. B. Kahng, R. Liang, H. Ren, S. S. Sapatnekar, and **B.-Y. Wu***, "**OpenROAD and CircuitOps: Infrastructure for ML EDA Research and Education**", *in Proc. VTS*, 2024. **(Primary Author)** [GitHub]

B.-Y. Wu, S.-Y. Fang, H.-W. Chang, and P. Wei, "**SpeedER: A Supervised Encoder-Decoder Driven Engine for Effective Resistance Estimation of Power Delivery Networks**", *in Proc. MLCAD*, 2022. **(Best Paper Award)**

Work Experience

Arizona State University Aug. 2023 – Present
Graduate Research Assistant *Tempe, Arizona*

- Developed a large-scale LLM dataset for physical design tool (OpenROAD) interaction, built a domain-specific LLM to generate OpenROAD scripts and answer OpenROAD-related queries, and created a self-correcting agent that iteratively refines scripts using real tool feedback.
- Proposed a diffusion-based generative framework to generate synthetic circuit layout heatmaps and open-sourced 23K+ synthetic samples to address data scarcity in ML-driven physical design research.
- Investigated LLM-based automated design rule violation (DRV) fixing and developed benchmarking infrastructure to evaluate LLM-driven DRC methodologies.
- Served as Topic Chair for ICCAD 2024 CAD Contest (Problem C) and contributed to OpenROAD/CircuitOps infrastructure to advance ML-EDA research and education.

Synopsys Inc. Oct. 2021 – June 2022
Intern (Technical-Engineering) *Taipei, Taiwan*

- Researched a novel Machine Learning-based (ML) solution estimating the effective resistance of Power Delivery Networks in advanced VLSI designs to speed up runtime and improve the accuracy of ML-driven IR analysis tools.
- Responsible for designing the data pipeline, the ML model architecture, and the entire effective resistance estimation workflow.

Academia Sinica Jul. 2020 – Nov. 2020
Full-time Research Assistant at Computational Finance and Data Analytics Lab *Taipei, Taiwan*

- Conducted research on a novel Transformer Encoder-based model with financial number category awareness, designing new pre-training and fine-tuning tasks to enhance its performance.
- Developed an Online Loan Application Recommender System for E.SUN Commercial Bank, achieving nearly 300% performance improvement. Responsibilities included workflow design, model development, and building Python APIs for industrial deployment.

Skills

Programming Languages: C, C++, Python, Tcl
EDA Tools: OpenROAD, OpenSTA, Innovus, Genus, ICC2, Design Compiler, HSpice, Virtuoso, Calibre
AI Tools: Claude Code CLI, Codex CLI, Cursor CLI
Language Ability: Mandarin (Native), English (Fluent)

Awards

Qualcomm Innovation Fellowship North America (QIF) May 2026

MLCAD Student Travel Grant Sept. 2024

Ferdinand A. Stanchi Fellowship Aug. 2024

DAC Young Fellow Travel Grant Jun. 2024

MLCAD Student Travel Grant Sept. 2023

Fulton Fellows Fellowship Aug. 2023

Best Paper Award at MLCAD 2022 Sept. 2022

Professional Service

Reviewer Activities
ACM TODAES (2025×2), IEEE TCAD (2026×2)

Topic Chair of Problem C at 2024 ICCAD CAD Contest Oct. 2024
IEEE CEDA *Newark, New Jersey*

2024 ASP-DAC Tutorial Talk Jan. 2024
ACM SIGDA *Incheon, South Korea*